X-104
CORROSION INHIBITOR PACKAGE (VDU OH)
Tank 2 m3 + 2 x 100 % metering pumps

J-201
STEAM EJECTOR (20 -> 130 mbar(a))
load 3020 kg/h
motive MP steam 7852 kg/h
2 x 50 % parallel

E-202
1ST-STAGE EJECTOR INTERCONDENSER
6.9 MW, 549 m², TEMA AXS

J-202
STEAM EJECTOR (120 -> 350 mbar(a))
load 486 kg/h
motive MP steam 1074 kg/h
2 x 50 % parallel

P-205A/B
HOTWELL SOUR WATER
13 m³/h @ 51 m
3.7 kW, API 610 OH2

E-203
2ND-STAGE EJECTOR INTERCONDENSER
1.0 MW, 79 m², TEMA AXS

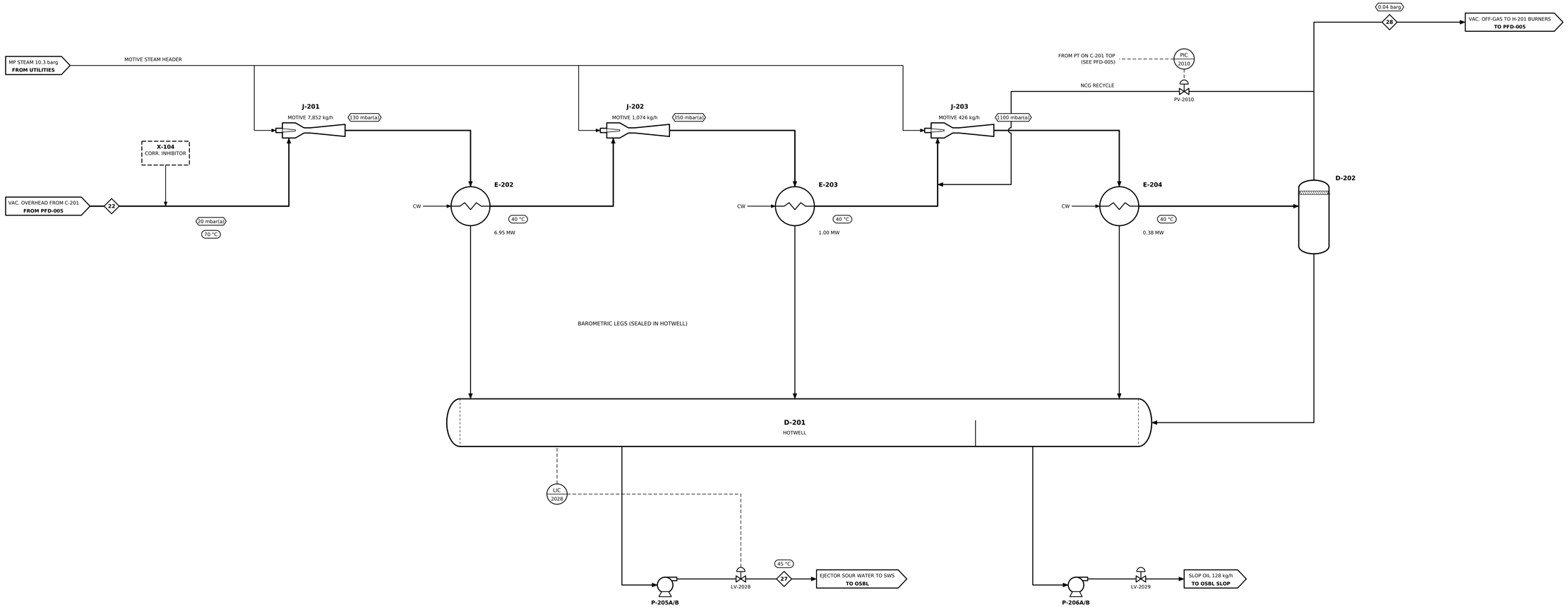
J-203
STEAM EJECTOR (330 -> 1100 mbar(a))
load 173 kg/h
motive MP steam 426 kg/h
2 x 50 % parallel

P-206A/B
HOTWELL SLOP OIL
1 m³/h @ 60 m
0.75 kW, API 610 API 674/675 PD

E-204
EJECTOR AFTERCONDENSER
0.4 MW, 30 m², TEMA AXS

D-202
VACUUM OFF-GAS KNOCK-OUT DRUM
ID 0.6 m x 2.0 m T/T

D-201
EJECTOR HOTWELL / SOUR WATER SEPARATOR
ID 1.3 m x 5.5 m T/T
boot slop oil compartment



STREAM No.	22	27	28
DESCRIPTION	Vacuum column overhead	Ejector sour water	Vacuum off-gas (NCG + air)
PHASE	VAPOUR	WATER	VAPOUR
TOTAL FLOW kg/h	3,020	12,141	102
TEMPERATURE °C	70	45	45
PRESSURE barg	-0.99	0.5	0.0
STD LIQ. FLOW BPSD	-	-	-
VAPOUR FRACTION (mass)	1	-	1
MOLECULAR WEIGHT	-	-	-

LEGEND

- 00 — H&MB STREAM No.
- 000 °C — TEMPERATURE
- 0.0 barg — PRESSURE
- TIC 000 — DCS CONTROLLER
- INSTRUMENT SIGNAL
- CONTROL VALVE
- SERVICE TO PFD-00X — OFF-SHEET CONNECTOR
- MAIN PROCESS
- SECONDARY PROCESS
- UTILITY
- FLUE GAS / AIR

NOTES

1. Each ejector stage 2 x 50 % parallel units; one shown. Motive MP steam.
2. Condensers E-202/203/204 drain via barometric legs (seal) to hotwell D-201.
3. C-201 top pressure PIC-2010 (transmitter on C-201 top, PFD-005) recycles off-gas to J-203 suction.
4. Stream numbers (diamonds) refer to H&MB CFU-000-PR-HMB-001; data from design case, normal operation.
5. Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PROCESS FLOW DIAGRAM
VACUUM EJECTOR SYSTEM & HOTWELL

DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	6 OF 6

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-200-PR-PFD-006**
REV: **A**